

Si-Xian Li

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EDUCATION

National Taiwan University Taipei, Taiwan

Bachelor of Science in Geography

Sep 2024 - Jun 2028

- Academic achievement: GPA: 4.03/4.3, Dean's List Award, Undergraduate Honors Program (in progress)
- Coursework: Cartography and Geographic Information System / Spatial Analysis / Data Analysis and Machine Learning / Programming / Data Structures / Remote Sensing / Statistics / Calculus / Probability

SKILLS

Languages & Libraries: Python / C++ / JavaScript / R / Pandas / NumPy / scikit-learn / XGBoost

Tools & Platforms: ArcGIS / QGIS / D3.js / deck.gl / MapLibre / Leaflet / GitHub Actions / NVIDIA OpenUSD

WORK EXPERIENCE

National Science and Technology Council

Taipei, Taiwan

Research Assistant

Apr 2026 – Present

Advisor: Chiao-Ling Kuo

- Built a geospatial text-mining pipeline applying NLP methods (CKIP named-entity recognition, LLM-assisted classification) to 10.7k+ social media posts, analyzing public discourse and urban dynamics around high-tech industrial park expansion; reduced data noise by 22%.
- Developed digital-twin workflows with NVIDIA OpenUSD, executing 3D urban simulations for data-driven planning analysis of industrial park expansion.

National Taiwan University

Taipei, Taiwan

Undergraduate Researcher

Sep 2025 – Present

Geospatial Computing Lab, Advisor: Tzai-Hung Wen

- Architected a spatiotemporal data processing pipeline for ~15,000 search query datasets (2016–2025), quantifying post-pandemic urban work-mode transitions across Taiwan and India.
- Co-authored international comparative analyses with Indian researchers to evaluate spatiotemporal variance in regional human mobility and work behaviors.

Academia Sinica

Taipei, Taiwan

Research Assistant

Sep 2025 - Present

depositar lab, Institute of Information Science, Advisor: Tyng-Ruey Chuang

- Co-developed an automated audio-visual data processing tool with a peer intern, standardizing raw data ingestion across the team and eliminating cross-member data formatting inconsistencies.
- Executed multi-site field research across Penghu to collect raw environmental audio-visual data using acoustic equipment, building the foundational dataset for the lab's soundscape platform.

Summer Intern

Jul 2025 - Aug 2025

Geographic Information System Center, Advisor: Tyng-Ruey Chuang

- Combined historical GIS and fieldwork in the Smangus Atayal community, georeferencing ~100-year-old Japanese colonial maps alongside local cultural data to build an interactive web map for spatio-temporal analysis.

PROJECT EXPERIENCE

An Analysis of Public Transport Accessibility in Kaohsiung City

May 2023 - Oct 2023

- Awarded 1st Prize, National High School Geographic Information System Competition in Taiwan
- Built a citywide PTAL accessibility model across Kaohsiung's 38 districts and 150 bus routes, metro, light rail, 56,129 reference points, adapting Transport for London's methodology to quantify transit equity and evaluate rail investment outcomes.

OTHER ACTIVITIES

NTU Varsity Track and Field team

Sep 2025 - Present

- Specialty: 400m (PB: 52.80) / 800m (PB: 02:07.15)
- Treasurer: Manage team finances and operations, including budgeting, dues collection, reimbursements for the team.

Academic Division, NTU Geography Student Association

Jul 2025 - July 2026

- Coordinated departmental events, including career development workshops, the NTU Azalea Festival (Department Expo).